HYDROWORX

YOUR POTENTIAL ◆ ACHIEVED

2000 Series™

Operating and Maintenance Manual

Disclaimer and Confidentiality Notice

This confidential Operating and Maintenance Manual (the "Manual") is being delivered to parties (the "Recipient") who have purchased a HydroWorx Pool from HydroWorx International, Inc. This Manual is highly confidential and proprietary in nature and is submitted for use by Recipient solely in connection with Recipient's consideration of the enclosed information. The use of this Manual for any other purpose is not authorized.

By accepting this Manual, the Recipient agrees to keep confidential the information contained herein or made available by the Company in connection with any service, maintenance, or repair of the HydroWorx Pool, to make no copies of the Manual in whole, in part, or in any form and to use this Manual solely for the purposes of understanding the proper use of the HydroWorx Pool.

2000 SERIES
OPERATION MANUAL INDEX

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Overview of Operation
&
Maintenance Manual

HydroWorx 2000 Series

Your HydroWorx 2000 Series has been custom fabricated for you according to your particular specifications and will provide you years of reliable service and therapeutic treatment if maintained properly.

Like any mechanical device, your HydroWorx Pool must regularly be inspected and have regular preventative maintenance procedures performed on the pool.

You can purchase an additional year or years of comprehensive or limited maintenance and service coverage for your pool from HydroWorx.

The information contained within this manual will profile various components, systems and parts integrated into your HydroWorx 2000 Series Pool. These profiles are for information purposes only and are not to be used for repair or replacement unless the repair or replacement activity has been specifically authorized by HydroWorx. Any unauthorized repair or replacement activity will void your limited warranty and may cause harm or injury to both the product and user.

In addition to the profile data, you will find diagrams, procedures, instructions, and other basic data for your reference.

HydroWorx 2000 Series Pool

Limited Warranty

HydroWorx International, Inc. (HydroWorx) warrants solely to the original Buyer that this product, consisting of the pool shell, integrated treadmill, jet system(s), electronic controls and water circulation equipment will be free from defects in materials and workmanship, when given normal, proper and intended installation and usage, for the period set forth below from the date noted on signed start-up completion form.

LIMITED WARRANTY PERIOD	COMPONENTS COVERED
One (1) Year	▪ Water Circulation Equipment (heater, filter housing, brominator and pump)* ▪ Badu Jets and Pump* ▪ Cluster Side Jets* ▪ Treadmill Motor and Gearbox* ▪ Treadmill Seal and Bearings* ▪ Lift Seals and Bearings* ▪ Treadmill System* ▪ Treadmill Belt* ▪ Lift System* ▪ Control System ▪ Computer System ▪ Underwater Video System
Five (5) Years	Tank PVC Membrane Liner* Structural Integrity of ▪ Walls ▪ Lift* ▪ Treadmill Frame*

***Proper maintenance and lubrication schedules must be followed to keep the warranty intact.**

During the limited warranty period, HydroWorx will repair or replace, at its option and without cost to the Buyer, any portion of the product which is confirmed by HydroWorx to be defective.

This warranty shall not apply to any expendable component, such as, but not limited to chemicals, fuses, filters, and lubricants. HydroWorx shall have no obligations to make repairs or replacements which result, in whole or in part, from (I) catastrophe, fault or negligence of Buyer, (II) improper or unauthorized use of product, (III) maintenance or installation of the product which is not in full compliance with the requirements of the HydroWorx Operating Manual and Software Manual, (IV) the presence of debris or silt in the product, (V) the failure to maintain water cleanliness, chemistry and temperature in

accordance with the recommendations of circulation equipment suppliers, (VI) any modification of the product (including, but not limited to, removal or affixing any object to the equipment or removing any component part of the product without prior specific written instruction from HydroWorx), or (VII) causes external to the product such as, but not limited to, power failure.

In the event after inspection by HydroWorx the product proves not to be defective or is not subject to the warranty protection because of normal use, wear and tear, misuse, lapse of time, accident, modification, or any other exclusion specified above, the Buyer will be charged for repair or replacement services at HydroWorx's standard rates then in effect together with the "portal to portal" expenses incurred by HydroWorx in providing the service.

THE PROVISIONS OF THE FOREGOING WARRANTY ARE IN LIEU OF ANY OTHER WARRANTY, WHETHER EXPRESS OR IMPLIED, WRITTEN OR ORAL, INCLUDING ANY WARRANTY OF MECHANTIBILTY OR FITNESS FOR A PARTICULAR PURPOSE. HYDROWORX'S LIABILITY ARISING OUT OF THE MANUFACTURE, SALE OR SUPPLYING OF THE PRODUCTS OR ITS USE OR DISPOSITION WHETHER BASED UPON WARRANTY, CONTRACT, TORT OR OTHERWISE, SHALL NOT EXCEED THE ACTUAL PURCHASE PRICE PAID BY THE BUYER FOR THE PRODUCT. IN NO EVENT SHALL HYDROWORX BE LIABLE TO BUYER OR ANY OTHER PERSON OR ENTITY INCLUDING, BUT NOT LIMITED TO, LOSS OF PROFITS OR LOSS OF USE OF DAMAGES ARISING OUT OF THE MANUFACTURE, SALE OR SUPPLYING OF THE PRODUCT THE FOREGOING WARRANTY APPLIES TO THE BUYER ONLY AND SHALL NOT ETEND TO ANY OTHER PERSON OR ENTITY, INCLUDING, WITHOUT LIMITATION, CUSTOMERS OF THE BUYER.

SOME STATES DO NOT ALLOW LIMITATIONS ON HOW LONG AN IMPLIED WARRANTY LASTS, OR THE EXCLUSION OR LIMITATION OF INCIDENTAL OR CONSEQUENTIAL DAMAGES, SO THE ABOVE LIMITATION OR EXCLUSION MAY NOT APPLY TO SOME BUYERS. THIS WARRANTY GIVES THE BUYER SPECIFIC LEGAL RIGHTS, AND THE BUYER MAY ALSO HAVE OTHER RIGHTS WHICH VARY FROM STATE TO STATE.

WARNING: USER OR PATIENT SHOULD NOT USE THIS EQUIPMENT WITHOUT CONSTANT ADULT SUPERVISION FROM A QUALIFIED PRACTITIONER WHO HAS BEEN TRAINED ON THE USE OF THE HYDROWORX SYSTEM. PATIENTS OR USERS SHOULD CONSULT HIS/HER PHYSICIAN BEFORE USING THIS EQUIPMENT. AT ALL TIMES HYDROWORX DISCLAIMS AND SHALL NOT BE LIABLE FOR ANY PERSONAL INJURY SUSTAINED BY ANY USER WHETHER AUTHORIZED OR UNAUTHORIZED, OF THE PRODUCT.

2000 Series Pool Usage Guidelines and Warnings

WARNING

Never use the HydroWorx pool without supervision from a qualified aquatic therapy practitioner or exercise specialist.

WARNING

Never dive or jump into your HydroWorx pool. Any unauthorized or extreme use may result in injury or death.

WARNING

Always have patients consult their physician or other medical professional before utilizing the HydroWorx pool.

WARNING

Unauthorized service, repair or replacement of parts will void your warranty and may damage your HydroWorx pool or injure your patients or qualified practitioner.

WARNING

Never operate the treadmill with more than 4 people at one time on treadmill floor.

WARNING

Never operate the treadmill without water or in water less than 3 feet of depth.

WARNING

Never raise or lower the treadmill floor when the treadmill is engaged and spinning.

WARNING

Never operate the treadmill with debris or other objects in the pool as these objects may damage the safe and smooth operation of the treadmill.

WARNING

If shoes are worn on the treadmill, they must not have shoelaces that may make contact with the treadmill.

WARNING

Always have patients stand in the middle of the treadmill floor when engaging the lift to lower or raise the treadmill floor.

WARNING

Always have patients walk, jog, or run in the middle of the treadmill floor to avoid foot contact with the treadmill side and back stainless steel cover plate.

Electrical Components Installation & Maintenance Inspection Checklists and Warranty Activation Forms

Please reference the electrical installation checklists included in this operations manual for items to be completed by the projects qualified licensed electrician upon installation and recommended follow-up yearly inspections. ***These checklists are critical to ensure installation meets the proper specifications of all referenced manufacturers and the owner should request copies of all completed forms from the electrical contractor before allowing the HydroWorx pool(s) to be used. At a minimum, yearly inspections should be performed by a qualified licensed electrician to re-inspect all items noted on the checklists to ensure ongoing compliance.***

The documents included in this operations manual are the following:

Delta Warranty Activation Form

- This is the form that comes with every Delta electrical panel that must be completed and faxed or mailed to Delta to activate warranty on the electrical panel and confirm the unit is installed per the original manufacturer's guidelines. This is the main power panel that controls functions on the HydroWorx pool on the following pool models: 2000, 1200

Coates Heaters Installation & Maintenance Inspections Checklist

- This form must be completed to confirm the unit is installed per the original manufacturer's guidelines. This is the unit that provides heat to the HydroWorx pool on the following pool models: 2000, 1200, 3500, 3500i, 750, 750i, 500, 500i, Instafit, L-Series, T-Series, and Thermal Plunge Series.

Speck Swim Jet Systems & Equipment Packs Electrical Inspection Checklist

- This form must be completed to confirm the unit(s) is installed per the original manufacturer's guidelines. Speck equipment is commonly part of the jet systems and circulation systems on the following pool models: 2000, 1200, 3500, 3500i, 750, 750i, 500, 500i, Instafit, L-Series, T-Series, and Thermal Plunge Series.

IMPORTANT INFORMATION

HydroWorx
1200/2000 Series

HMI Control System
Owner's Manual and Installation Guide

PLEASE READ BEFORE INSTALLING OR SERVICING THIS DEVICE

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HydroWorx 1200/2000 Warranty Activation

THIS SHEET MUST BE COMPLETED AND SIGNED BY QUALIFIED, LICENCED ELECTRICIAN AND FAXED OR MAILED TO ISTECH TO ACTIVATE WARRANTY ON THIS SYSTEM.

Please initial all boxes when completed.

INITIAL	DESCRIPTION OF CHECK
	All wiring of proper gauge stranded 75* CU wire specified on print.
	All motors grounded with separate wire of same gauge as motor wiring or greater, as specified on print.
	All wiring connections tightened to specified torque.
	Each motor controlled by a VFD has wiring run in separate conduit.
	All entry points in enclosure sealed with UL Listed Type 12 Hub/ Fitting and/or sealing Device.
	All DC cabling (HMI, Communication, Sensors, Camswitch) is at least 8" from AC wiring.
	All wire connectors are dry & watertight.
	Power feed (including Ground) is of proper gauge stranded 75*C CU wire specified on print.
	Power feed is of proper voltage and protected by suitable circuit breaker.
	GFI Test button(s) are operational.
	Enclosure is properly mounted according to NEC & UL508 regulations, in a non-corrosive indoor environment not exposed to freezing temperatures.
	Area in front of Enclosure is free and clear of obstruction according to NEC and UL508 regulations.

Panel Serial Number:	
Panel UL Listing sticker number:	
Panel Customer Name:	
Panel Customer Location:	
Date Checklist Completed:	
Completed By (print Name):	
Completed By (sign Name):	
Phone Number:	

Please mail or fax to : **ISTECH, Incorporated**
4691 Raycom Rd
Dover, PA 17347
Phone : (717) 764-5565
Fax: (717) 467-5780

1. Introduction

1.1 Products Covered in the Manual

This manual covers the control systems manufactured by ISTECH, Inc. for the HydroWorx 1200 Series through 2000 series swimming pools.

1.2 Product Changes

ISTECH, Inc. reserves the right to discontinue or make modifications of its products without prior notice and hold no obligation to make modifications to previously sold products. ISTECH, Inc. holds no liability for losses of any kind which may result from this action.

1.3 Limited Warranty

Seller warrantees the machinery or equipment to be delivered against defects in material and workmanship, under the normal use and service for which it was designed, for a period of ONE (1) YEAR after date of delivery to customer's facility. Any alteration to the components for the equipment performed by the buyer other than specified by this manual shall void this warranty. Seller's obligation under this warranty is limited to the repair or replacement of material or the correction of any workmanship shown to seller's satisfaction to be defective. Buyer must notify the seller in writing immediately of any and all such defects. This limited warranty does not extend to components not manufactured by seller, but the manufacturer's warranty will be passed through seller to buyer. THE REPLACEMENT OF NORMAL WEAR COMPONENTS IS NOT INCLUDED IN THIS WARRANTY.

1.4 Limitation of Liability

Seller's liability with respect to the machinery or equipment sold shall be limited to a correction of the defects as defined in the LIMITED WARRANTY described above. Seller's liability under, for breach of, or arising out of this agreement and/or sale shall be limited to refund of the purchase price. In no event will the seller be liable for costs to procure substitute goods by the buyer. In no event will the seller be liable for any special, consequential, incidental, or other damages (including, without limitation, loss of profit, sales, use, time and data) whether for breach or repudiation of contract, breach or warranty, negligence or otherwise. This exclusion also includes any liability that may arise out or third part claims against buyer.

1.5 Receiving

Inspect all cartons for damage, which may have occurred during shipping. Carefully unpack all equipment and inspect thoroughly for damage or shortage. Report all damage to carrier and/or shortages to supplier. All major components and connections should be inspected for damage and tightness, with special attention to PC boards, plugs, knobs, switches, terminals, and terminal block jumpers. Complete the Product Warranty Registration and fax to ISTECH.

Upon unpacking, please check small red "Drop 'n' Tell" devices inside control cabinet. If any of these devices is showing a BLUE arrow, the unit has exceeded 25 Gs and the supplier must be contacted immediately.



2. Installation

2.1 Safety Instructions

1. Read and FOLLOW all instructions.
2. ⚠ **WARNING!** Do not allow children to operate this product unless closely supervised by an adult at all times.
3. ⚠ Before installing or servicing this system, be certain all power supplies are shut "OFF" to avoid risk of electrical shock or death.
4. ⚠ This system MUST be protected by a Class A Ground Fault Interrupter - intended for personnel protection. Disabling, overriding, or any other means of altering the installed GFI device will cause risk of electric shock or death.
5. This equipment is designed for a 120/208-Wye configuration mains power feed, with Neutral and Ground. If other power configurations are required, contact ISTECH for further instructions.
6. DO NOT bond the Neutral terminal bar to Ground inside the main enclosure except at the mains terminals. This will cause the GFI to trip. If it is necessary to bond the Neutral to Ground at the control panel, do so at the mains terminals at the top of the panel beside the disconnect switch.

7. ⚠ All pool and spa equipment must be grounded with a bonding wire. A bonding wire is a solid 8-gauge wire that is connected to a lug on the exterior of each appliance in the equipment package. No conduit is needed because current is not normally carried by this wire. The gauge of the ground wire of any appliance **must be as large as the hot wire(s)** so it is capable of efficiently conducting electricity away from the appliance in the event of a short circuit or stray current, as well as prevent GFCI nuisance tripping.
8. ⚠ ISTECH recommends that **the main ground wire must be as large as the largest hot (current carrying) wire in use,** so it is capable of efficiently conducting electricity away from the appliance in the event of a short circuit or stray current, and to prevent GFCI nuisance tripping. All pool and spa equipment must be grounded. Current NEC does allow for a smaller ground conductor, however ISTECH recommends running a **full size ground conductor** for the safety of clients, and reduced GFI nuisance tripping. Please follow the electrical schematic and follow all applicable NEC guidelines for wire size.
9. All motor and device wiring should exit bottom of enclosure - near terminal connections.
10. ALL conduit entry points must use UL Listed **Type 12** Hub/Fitting and/or sealing device, to maintain the Type 12 Rating of the electrical enclosure.
11. Metal conduit is recommended to reduce motor noise and GFI nuisance tripping.
12. Treadmill motor, and Swim jet motor wiring MUST each be run in separate conduit from all other devices. Mixing other wiring in conduit may cause GFI nuisance tripping.
13. Never run motor wiring in the grey area (refer to fig.1). Grey area (fig.1) is designated for DC control. Running AC wiring in this area may cause system malfunctions and/or GFI nuisance tripping.
14. Use at least 75°C stranded copper MTW or THHN wire of proper size (refer to electrical schematic) when installing field wiring and power feed - DO NOT use solid wire or Romex.
15. ⚠ Do not run the HMI interface or Communications cable near motor wiring or other AC lines. Close proximity to AC lines may cause communications loss, and/or system malfunction.
16. After installation is complete, remove plastic cover from top air vent on the VFDs (variable frequency drives).

17. ⚠ Failure to observe any of the above instructions may cause system malfunction and/or GFI nuisance tripping.
18. Observe all NEC and local regulations when installing this system.
19. ⚠ High Voltage can be present for 5 minutes after system power has been disconnected. Risk of electric shock or death.
20. This control panel is a listed UL508A, Industrial control panel – refer to UL sticker on inside panel.
21. Save these instructions for future reference

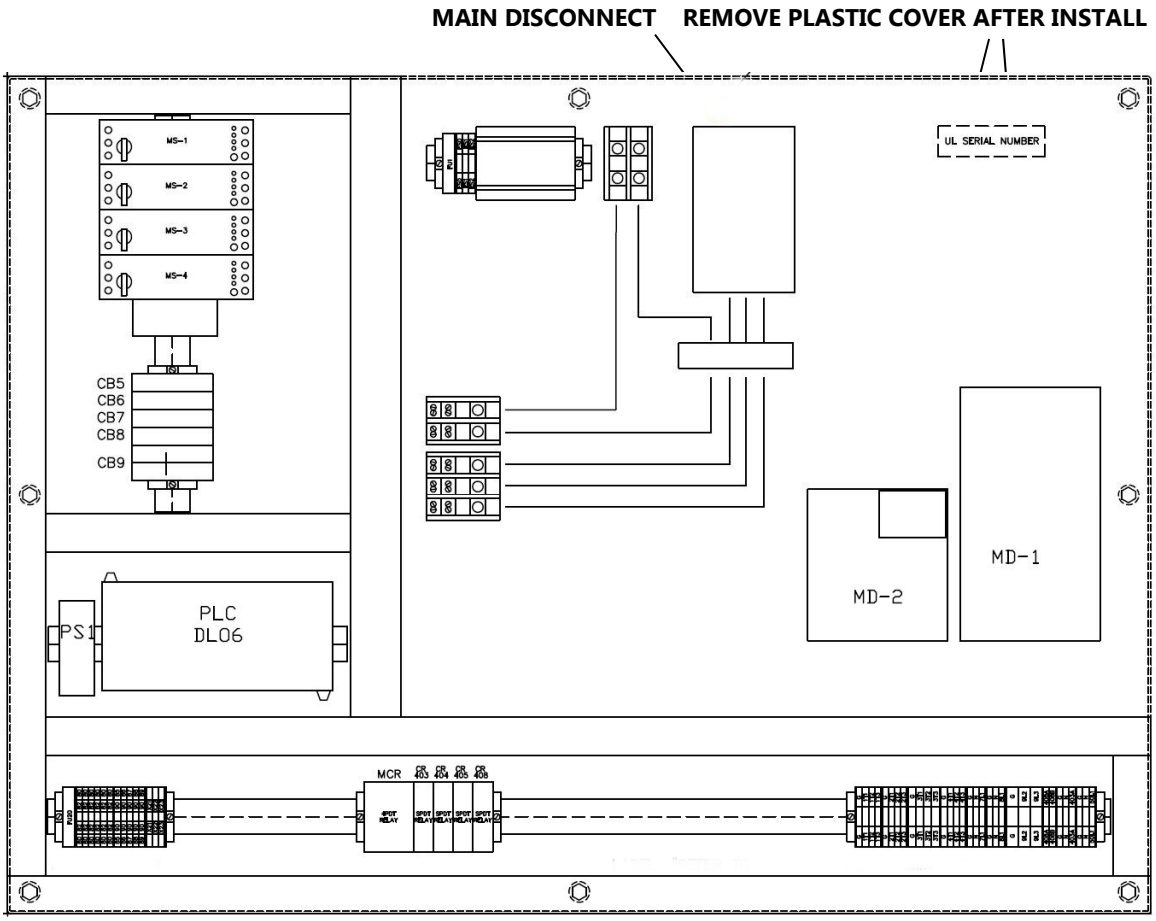
2.2 Mounting

This electrical control cabinet must be vertically mounted, observing all applicable NEC and local regulations. Mounting legs are provided on bottom of enclosure for models1200, 2000, and should be securely fastened to the floor. The electrical enclosure is UL **Type 5** and any installation modifications must be made so as to maintain the **Type 5** rating.

-FOR INDOOR USE ONLY-

⚠**WARNING:** THE UNIT MUST **NOT** BE INSTALLED WHERE SUBJECTED TO ADVERSE ENVIRONMENTAL CONDITIONS SUCH AS: COMBUSTABLE, OILY, OR HAZARDOUS VAPORS OR DUST; EXCESSIVE MOISTURE OR DIRT; STRONG VIBRATION; EXCESSIVE AMBIENT TEMPERATURES. CONSULT ISTECH FOR MORE INFORMATION ON THE SUITABILITY OF INSTALLATION IN A PARTICULAR ENVIRONMENT.

Fig. 1 – 1200 Series Panel (2000 Series similar)



2.3 Wiring Motors

For best results, and least chance of GFI nuisance tripping, ISTECH recommends using separate metal conduits for each motor or other device connected to the control panel. Use only stranded CU 75*C MTW or THHN wire of proper size (refer to electrical schematic) to decrease noise and prevent GFI nuisance tripping. Ground wire should be of same size as "hot" motor wires to decrease chances of GFI nuisance tripping. All NEC and local regulations must be observed when wiring the motors.

⚠ **MOTOR WIRING MUST EXIT ENCLOSURE BOTTOM TOWARDS THE RIGHT SIDE! RUNNING THIS MOTOR WIRING THROUGH THE LEFT SIDE WILL CREATE NOISE IN THE DC WIRING AND MAY CAUSE GFI NUISANCE TRIPPING, AS WELL AS SYSTEM MALFUNCTION.**

2.4 Wiring HMI Interface

The HMI interface in a low voltage (12V) device. The multiconductor cable connects via CPC connectors. The pigtail enters the lower left of the control panel and wire to the 2-level terminal blocks provided. The color coding is included on the electrical schematic. PLEASE call ISTECH if you are not sure how to connect this wiring.

⚠ CONNECTING THE HMI OR COMMUNICATION CABLE IMPROPERLY COULD DAMAGE THE INTERFACE AND/OR THE CONTROLLER!

2.5 CAMSWITCH Wiring (1200/2000)

Terminals are provided for connection to an 8- point camswitch to detect floor level. These switches are wired 'normally closed' for failsafe operation. To aid in setting up the camswitch, an override jumper is installed at the factory to allow free movement of the floor during installation. Please remove this jumper after setting up the camswitch. A second override jumper is provided to disable Automatic Movement of the Lift. Once the camswitch is setup, verified and tested, this jumper can be removed to enable Automatic Movement of the Lift. Refer to electrical schematic.

⚠**WARNING: FAILURE TO REMOVE THE SAFETY OVERIDE JUMPER MAY CAUSE SERIOUS INJURY OR DEATH. PERSONS SHOULD NEVER TRY TO USE THE TREADMILL OR THE JETS WHILE THIS JUMPER IS STILL CONNECTED, AS THESE DEVICES WILL NOT TURN OFF WHEN THE FLOOR IS AT LESS THAN 2 FT. DEPTH**.

2.6 Heater Wiring

The 1200 Series provides a 60A circuit breaker to power one 11k electric heater. If a larger heater is required, it should be powered by a separate electrical service, with GFI protection. Additionally, terminals are provided for remote control of the heater, to shut the heater off when multiple jets and/or the treadmill are running. Refer to the manufacturer's instructions for connecting the remote control circuit.

The 30Kwatt Heater provided by HydroWorx for the 2000 should be powered by a separate electrical service, with GFI protection.

2.7 Wiring Precautions

ISTECH recommends periodic shutdown of the system to tighten ALL electrical terminals on the control panel. Loose connections lead to burnt wires and destroyed components. Connections not being made tight enough, or corrosion from environmental elements can occur in the connection itself. Both of these situations cause a high temperature condition to occur because of the electrical resistance that develops over time, due to oxidized copper connections. The most common occurrences of this will be found in spa control or electric heater main power connections, main power connections inside the remote cut-off switch, main power connections inside the main breaker box in the facility, and even the main power connections to the heater element itself.

3. VARIABLE FREQUENCY DRIVES

The Variable Frequency Drive(s) (VFD, Inverter) incorporated in this system operates 3 phase 200-240VAC motors. For proper operation, safety, and to reduce GFI nuisance tripping, numerous parameters must be adjusted from the factory default values.

Parameter	Description	1000/1020 All	1001/1021 All	1200 TM	1200 Jets	2000 TM	2000 HD TM	2000 Jets	2001 Jets
A1-01	Parameter Access	2	2	2	2	2	2	2	2
A1-02	Control Method	2	2	2	2	2	2	2	2
B1-01	Reference Selection	1	1	1	1	1	1	1	1
B1-04	Reverse Operation	1	1	1	1	1	1	1	1
B1-07	Local/Remote	1	1	1	1	1	1	1	1
B1-08	Run Command	1	1	1	1	1	1	1	1
B1-17	Operation Permission	1	1	1	1	1	1	1	1
C6-01	Duty Cycle	0	0	0	0	1	0	0	0
C6-02	Carrier Frequency Selection	000F	000F	000F	000F	000F	000F	000F	000F
C6-03	Carrier Frequency Upper Limit	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
E1-01	Input Voltage***	208	208	208	208	208	208	208	208
E1-04	Max Output Frequency	60	50	60	60	60	60	60	50
E1-05	Max Voltage***	208	208	208	208	208	208	208	208
E1-06	Base Frequency	60	50	60	60	60	60	60	50
H6-02	Pulse Train Input Scaling	-	-	-	-	-	-	-	-
H6-08	Pulse Train Min. Input	-	-	-	-	-	-	-	-
L1-01	Motor Protection Selection	3	3	3	3	3	3	3	3
L1-02	Motor Protection Time	5	5	5	5	5	5	5	5
A1-01	Parameter Access	0	0	0	0	0	0	0	0

4. Ground Fault Interrupter (GFI)

4.1 Description

This control panel incorporates a UL Listed, Class A, Personnel Protection GFI device. This device will fault at 6mA/20ms leakage current.

4.2 Testing

A test button is provided on the door of the enclosure. To test/reset the device, turn the TEST/RESET Switch to TEST, and the unit should lose power after a few seconds. To reset the GFI, turn the TEST/RESET Switch to RESET, and the GFI will reset after a few seconds. The main breaker needs to be turned to RESET then to ON to return to operation.

4.3 Finding Ground Faults

This GFI device has a display showing the leakage current present. If the reading is <2mA, try turning off one circuit breaker at a time to isolate the circuit causing the ground fault.

4.4 Common Problems

Common problems which cause nuisance tripping are:

- **Surge Protector Receptacles** – these receptacles are designed to drain excess current to ground in the event of a surge or spike. This will trip the GFI device.
- **Line Filter Devices** – These devices have capacitors from line to ground to decrease electrical noise. This leakage current will trip the GFI device.
- **Improper Grounding** – Improper grounding, especially bonding neutral and ground together on the load side of the GFI, will cause the GFI device to trip.
- **Pinched Insulation** – Pinched wire insulation causes resistance in the connection and may cause GFI nuisance tripping, especially with Neutral connections.
- **Shorted Heating Element**.
- **Shorted Motor**.
- **Burnt/Frayed wiring**.
- **Moisture** – in the enclosure, inside the conduits, or in cabling.
- **Bonding and grounding** – refer to Section 2.1 of this manual for proper bonding and grounding procedures.

5. System Operation

5.1 Powering Up the System

5.1.1 Turn on the main power at the disconnect switch (main enclosure).
5.1.2 Make sure all "EMERGENCY STOP" Pushbuttons are pulled out
5.1.3 Press the Amber "POWER RESET" Pushbutton on the Main Enclosure, or on the HMI Box.
5.1.3.1 The amber pushbutton should be illuminated now.
5.1.3.2 This condition will be referred to as "Powered Up"
5.1.4 If the Power will not come up, check to make sure the HMI cable (9 pin multiconductor cable) is plugged securely into the back of the pendant and into the pigtail cable at the main enclosure.

5.2 Testing the Ground Fault System

5.2.1 Turn the TEST/RESET Switch to Test for a few seconds on the main enclosure.

The main disconnect circuit breaker should trip.

⚠ **The GFI System should be tested frequently to ensure safe operation.**
IF THE GFI DOES NOT TRIP, CONTACT ISTECH. DO NOT LET ANYONE INTO THE WATER, SINCE THE POOL IS NOT ELECTRICALLY PROTECTED.

5.2.2 The GFI can be reset by disconnecting the mains feed power to the system, or by turning the TEST/RESET Switch to Reset for a few seconds

5.3 Automatic Water Level Control (1200/2000 Series)

The system is equipped with a water level control circuit (where installed). The system utilizes two optical sensors to detect the water level.

5.3.1 Manual Operation
5.3.1.1 Turn the Keyswitch on the main enclosure to the MAN position to add water to the pool.
5.3.1.2 Turn the Keyswitch back to the OFF position to stop filling.
Caution – You can overfill the pool using this method.

5.3.2 Automatic Operation

5.3.2.1 Press EMERGENCY STOP on the pendant or on the main electrical enclosure.

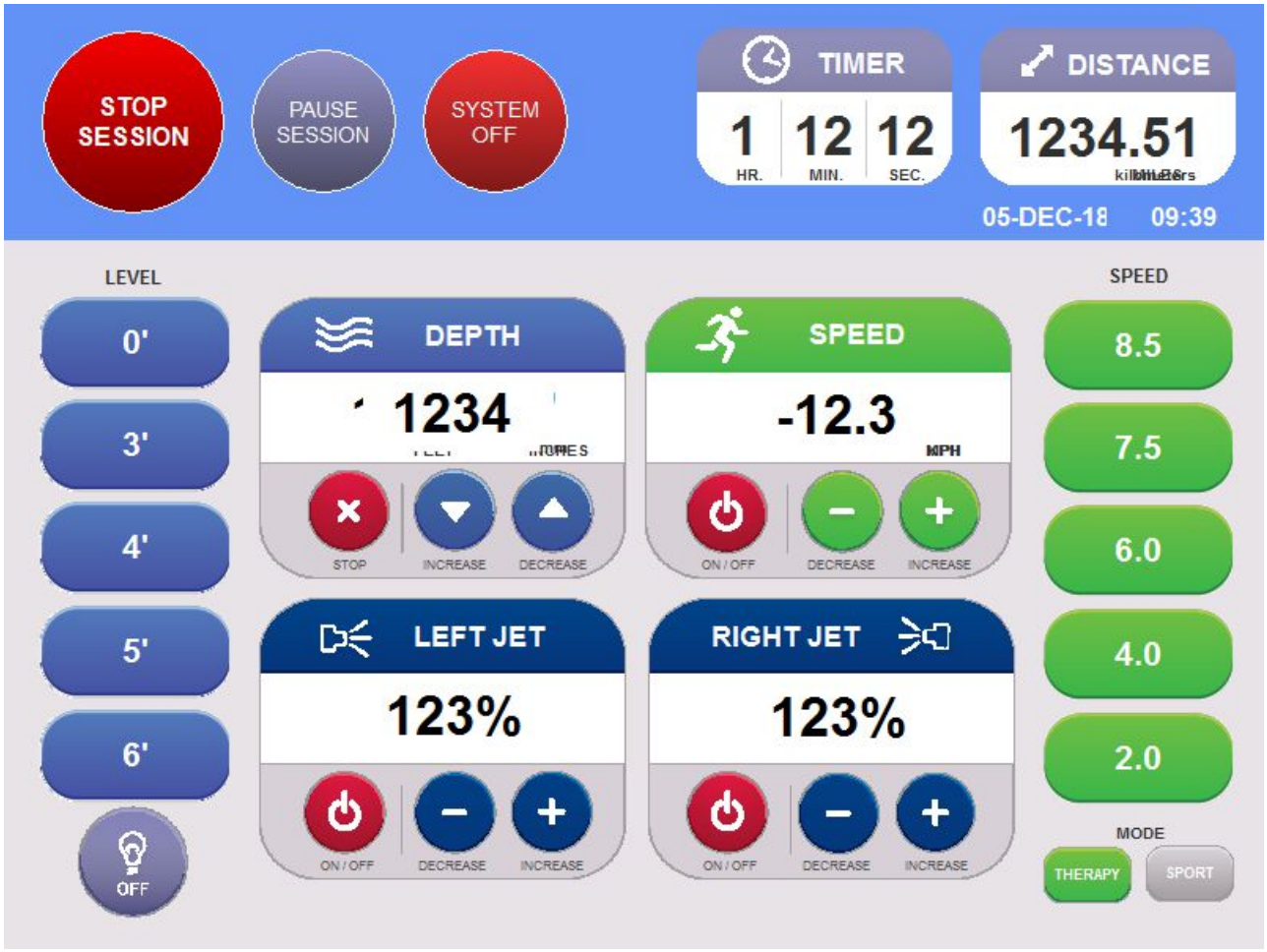
5.3.2.2 Turn the Water Fill Keyswitch to AUTO.

5.3.2.3 After 30 seconds (to allow the water to settle), when the water falls below the lower water sensor, the solenoid will open the valve, allowing water to flow into the pool until the upper water level sensor is reached. The Keyswitch will remain in the AUTO position until switched back.

Note: THE FLOOR LEVEL MUST BE AT 2 FT OR DEEPER TO USE AUTO WATER FILL. 0FT OR MAINTENANCE LEVEL WILL NOT PERMIT THE WATER TO FILL, TO PREVENT OVERFLOW WHEN THE FLOOR IS LOWERED BACK DOWN INTO THE WATER.

6. HMI Screens

6.1 Main Screen



Stop Session:
Press to STOP the Session. Treadmill and Jets will all STOP – Stops the Session Timer and Distance. Starting any moving feature will start a new session.

Pause Session:
Press to PAUSE the Session. Treadmill and Jets will all STOP Jets will remember speed Pauses the Session Timer and Distance. If PAUSE is pressed for 5 seconds, the Session Timer and Distance will be Reset.

SYSTEM OFF/ON
RED indicates that Control Power is OFF. GREEN indicates that Control Power is ON
All functions are disable with Control Power OFF. The Filter Pump and Heater remain operational.

Power Reset (Amber Lighted Pushbutton):
Press to turn Control Power ON - Amber Light turns ON

NOTE: If the HMI communications fail and Control Power is ON, Press and hold the POWER RESET for 5 seconds and the LIFT will move up toward the floor position as long as the Pushbutton is held. The LIFT will stop at the Floor Position. This feature will function even with the HMI online.

Emergency Stop (Red Mushroom Pushbutton):
Press to turn Control Power OFF
Twist the Mushroom Clockwise ¼ Turn to Release
Press POWER RESET to turn ON Control Power to return to operation

Cable Emergency Stop (At Back of Pool):
Pull Cable to turn Control Power OFF – Press POWER RESET to turn ON Control Power

Depth:
STOP
Press to Stop LIFT
INCREASE
Press to Jog LIFT Down
DECREASE
Press to Jog LIFT Up

Level:
Ovals turn Dark Blue when LIFT is at the indicated Depth

If you press an Oval indicating a Depth you would like the LIFT to move to, the Oval will Flash. Press the appropriate Depth INCREASE or DECREASE button to confirm the Automatic LIFT Move. You may Press DEPTH – STOP at any time to cancel the Move.

NOTE: Automatic LIFT functions are disabled in LEVEL SIMPLE MODE.

Light:
Press to Turn JET Light(s) ON – Control Power must be ON for LIGHT to operate

LEFT JET / RIGHT JET / FRONT JET:
ON/OFF – (Jets can operate 3' to 6')
Press to START or STOP Jet
DECREASE
Press to Decrease Jet Speed
INCREASE
Press to Increase Jet Speed

NOTE: If you press the Speed % Box, a number pad will appear were you can set the Jet Speed in %. This Speed is remembered unless the Session Stop Button or EMERGENCY STOP is pressed.

Treadmill:
ON/OFF
Press to START or STOP Treadmill
DECREASE
Press to Decrease Treadmill Speed
INCREASE
Press to Increase Treadmill Speed

Child Mode (Treadmill can operate 1' to 6')
Standard Mode (Treadmill can operate 2' to 6')

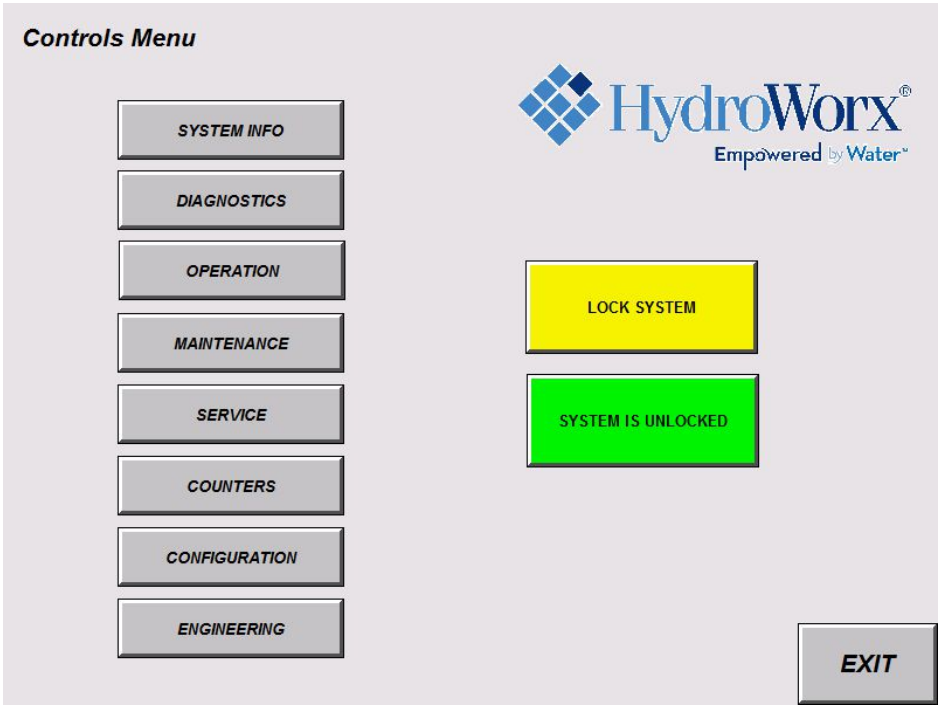
Speed:
Ovals turn Dark Green when Treadmill is at the indicated Speed

If you press an Oval indicating a Speed you would like the TM to ramp to, the Oval will Flash and the TM will ramp to the selected Speed.

Mode:
THERAPY - Displays Lower THERAPY Speed References
Press for 2 Seconds to change modes
SPORT - Displays Higher SPORT Speed References
Press for 2 Seconds to change modes

NOTE: **Do not use the speed controls to stop the Treadmill or the Swim Jets. Make sure you press OFF, STOP or PAUSE. If the Treadmill or Swim Jets are ON with Zero speed for some time, the drives may Fault.**

6.2 Controls Menu (Accessed by HIDDEN Button behind DISTANCE)

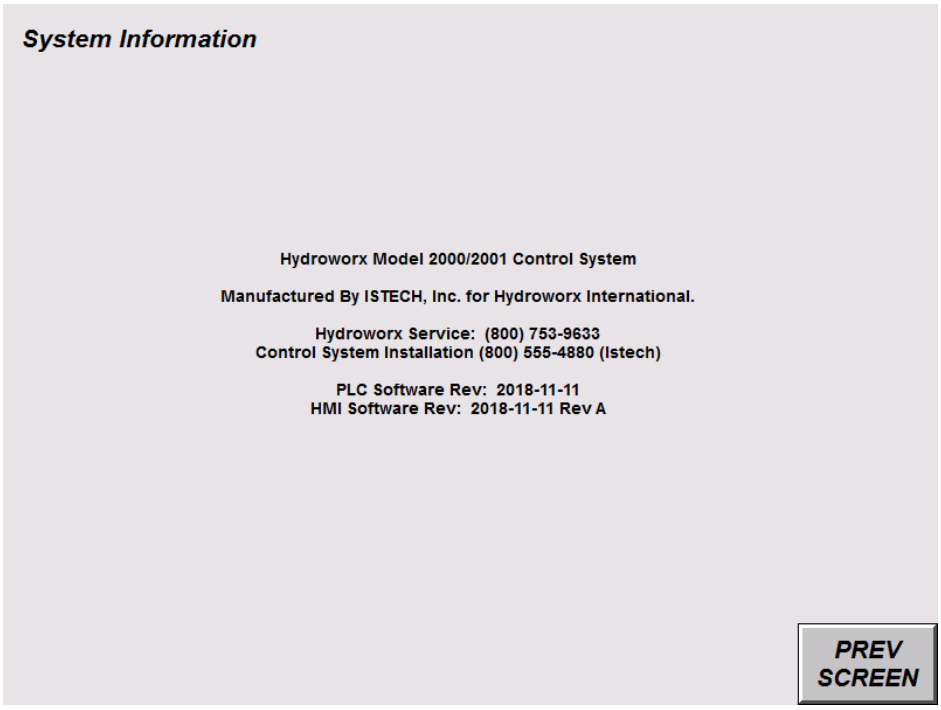


Lock System:
Press to Lock HMI Control Functions – Box will turn Red and say System Locked
Press to Unlock HMI Control Functions – Password Required – Box will turn Yellow

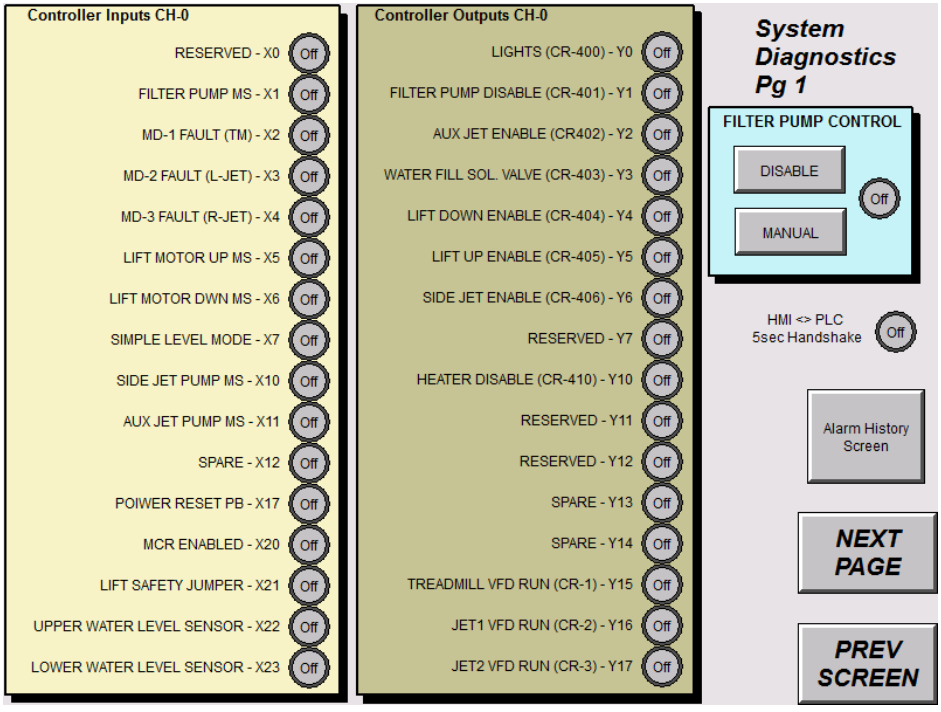
HW2000 Password: 2000

HW1200 Password: 1200

6.3 System Info Screen



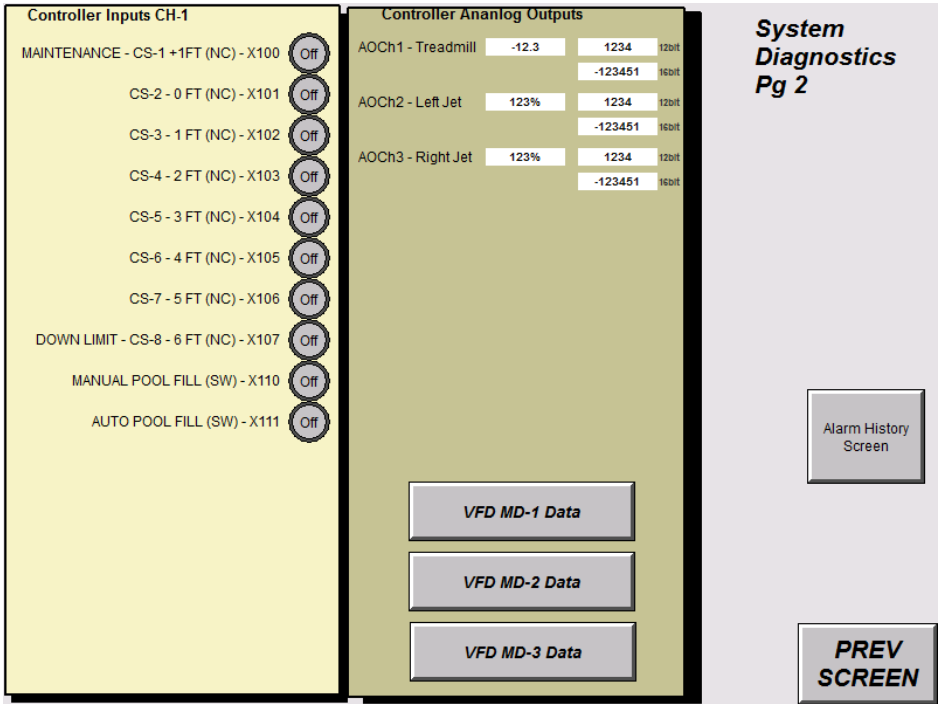
6.4 Diagnostic Screen Pg1



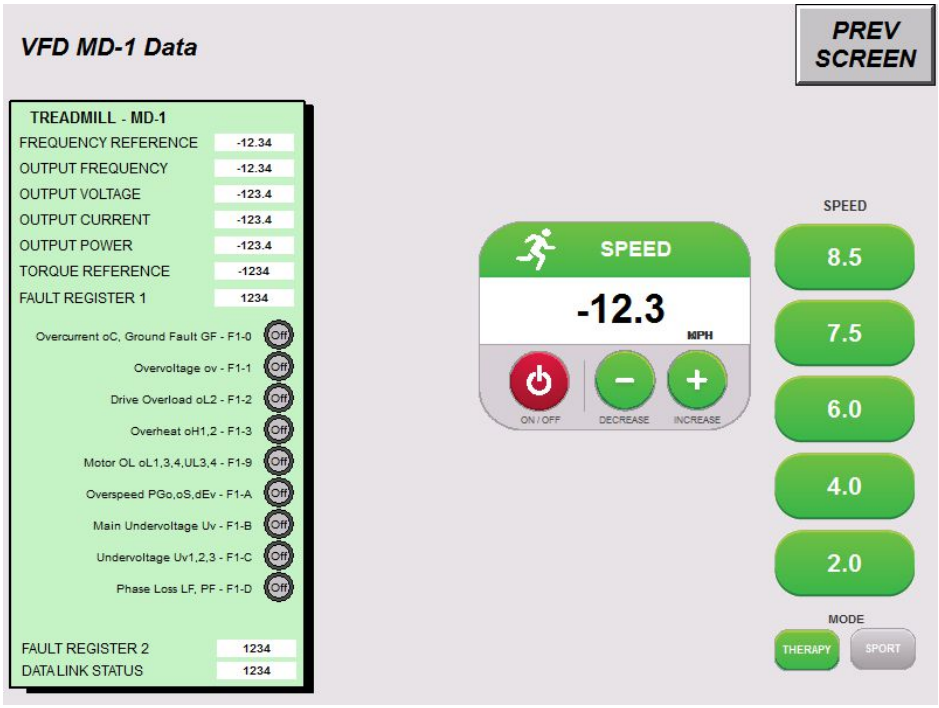
Filter Pump Control:
DISABLE – Press to Disable Filter Pump – Button will turn Yellow
MANUAL – Press to Select Auto programming for Filter Pump

NOTE: Filter Pump Auto programming is setup on the Configuration Screen

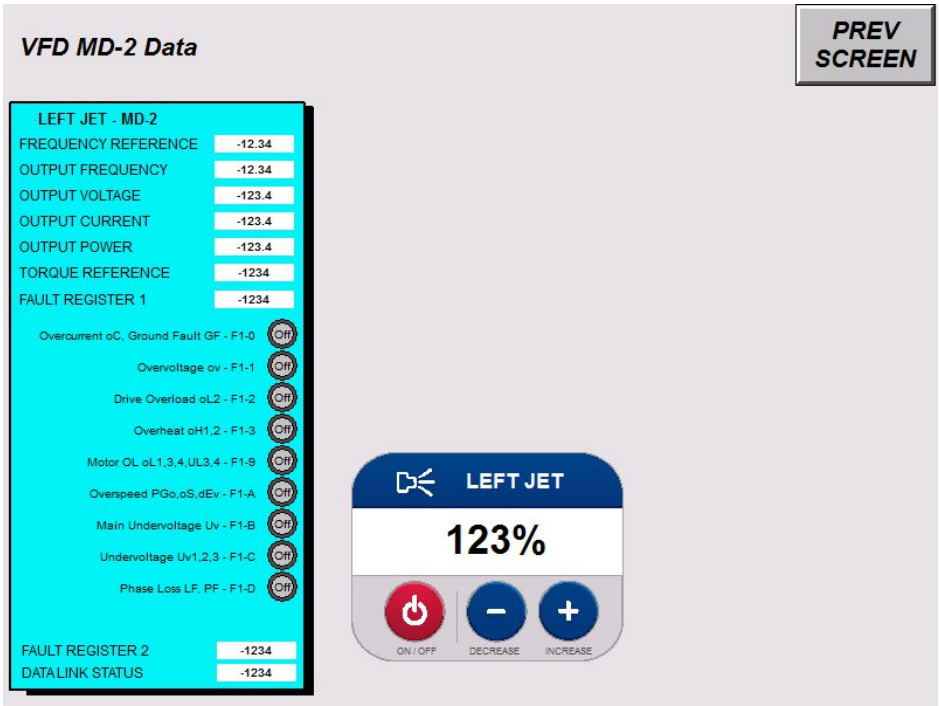
6.4.1 Diagnostic Screen Pg2



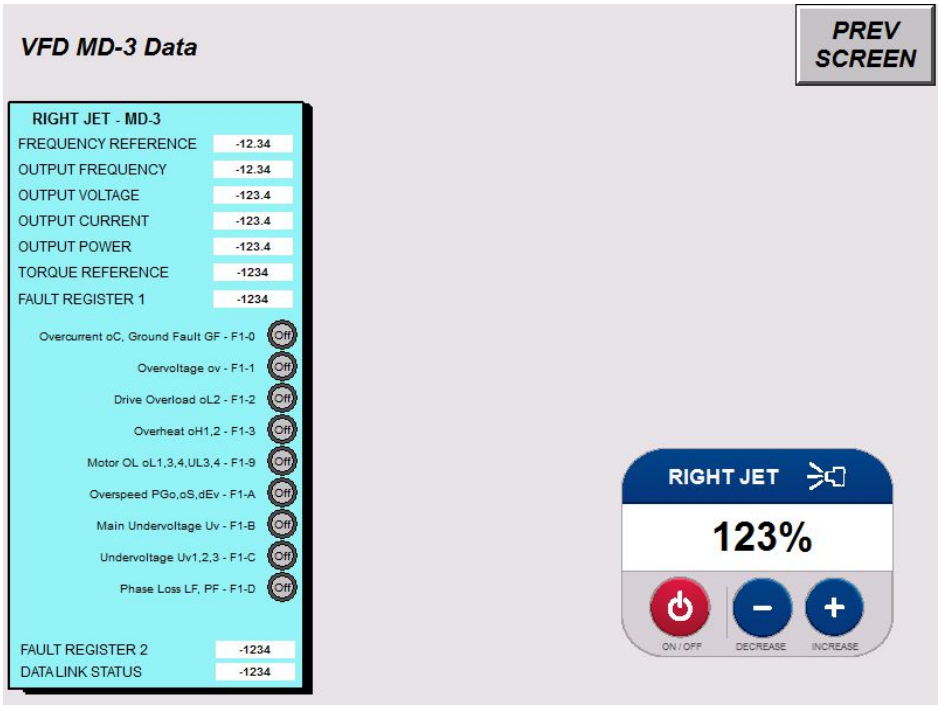
6.5 Diagnostic Screen – VFD MD-1 Data



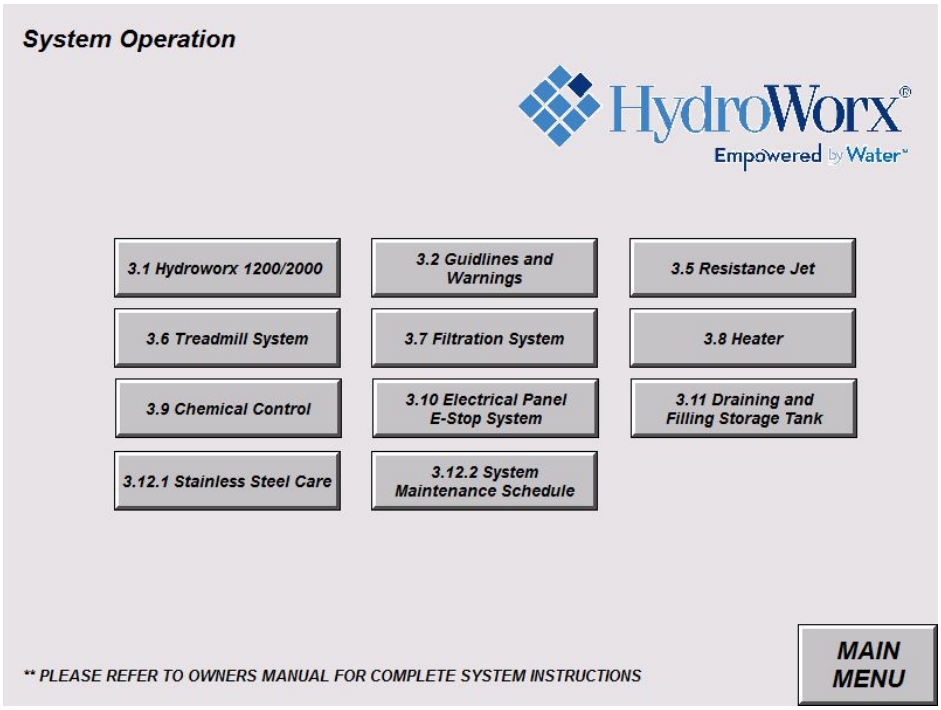
6.6 Diagnostic Screen – VFD MD-2 Data



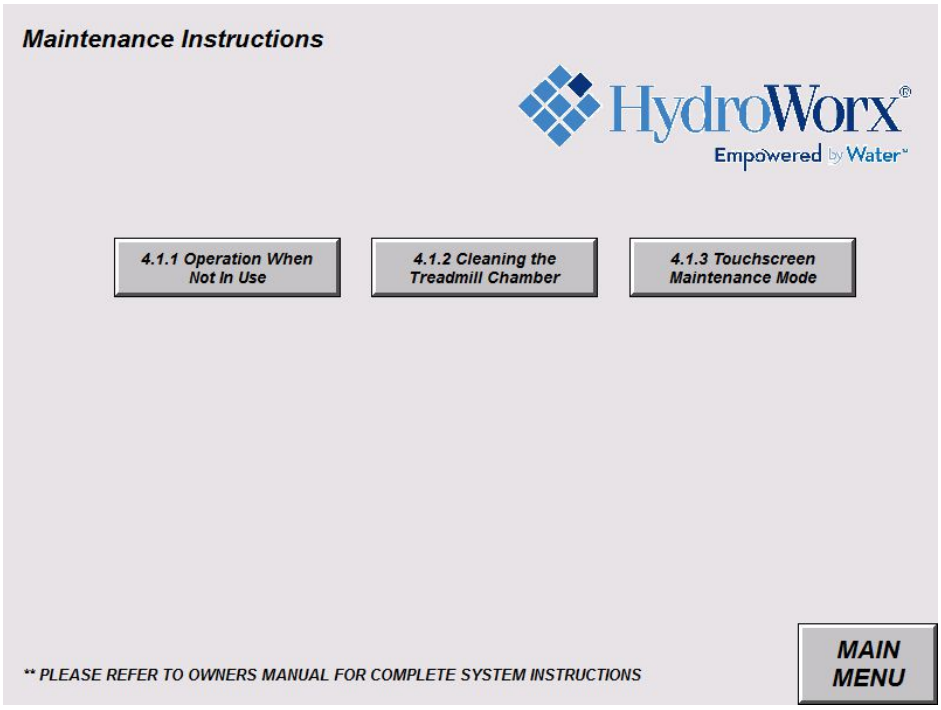
6.7 Diagnostic Screen – VFD MD-3 Data



6.8 System Operation Menu



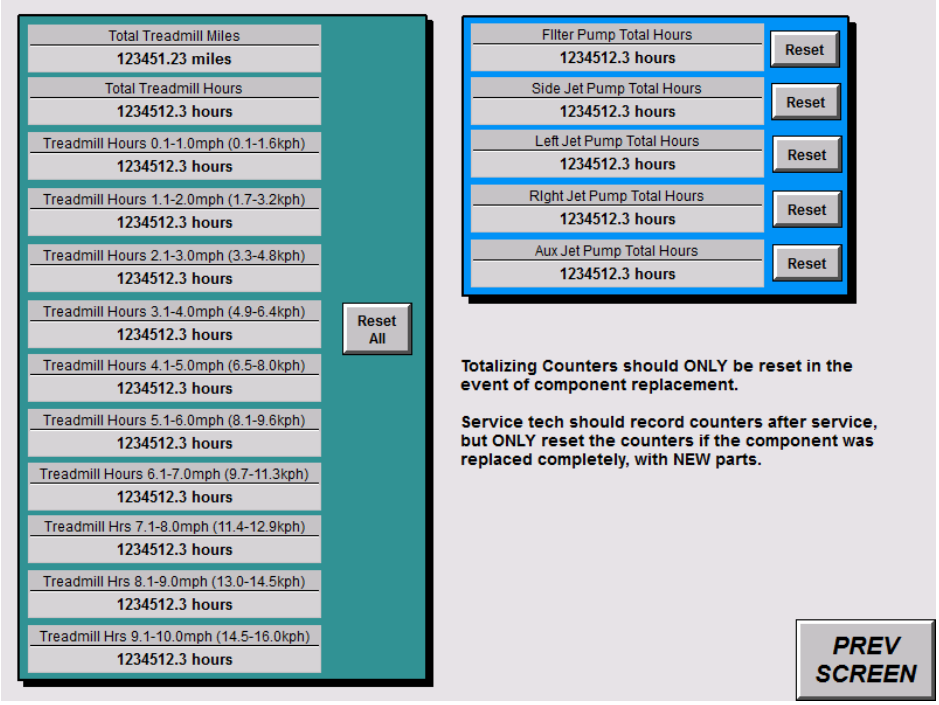
6.9 Maintenance Instructions Menu



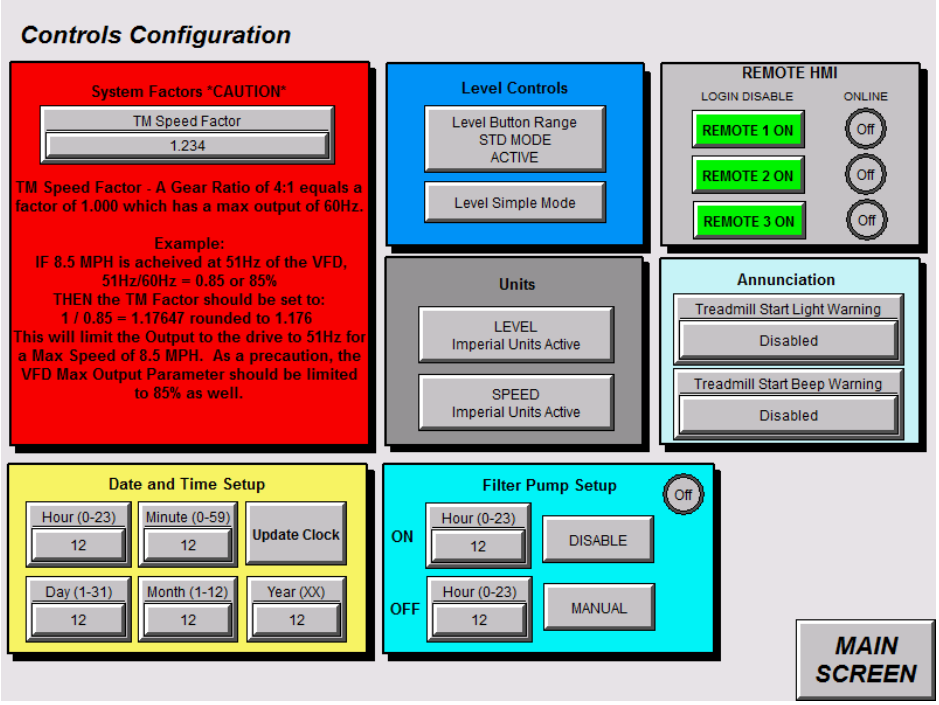
6.10 Service Information Screen



6.11 Counters Screen



6.12 Controls Configuration Screen – Password Required



***Password Required to Change Configuration**

TM Speed Factor:
Sets the Full Speed of the system to actual TM Belt Speed

Date and Time Setup:
Sets the Date and Time. Press Update Clock when complete.

Level Controls:
Level Button Range
Press to change Lift Level to Child Mode (Treadmill can operate 2' to 1')
Press to return to Lift Level to Standard Mode (Treadmill can operate 3' to 2')
Level Simple Mode – When Yellow, this indicates that the Lift Auto Level function is disabled

Units:
Level
Press to change Level to Metric Units, Press to return to Imperial Units
Speed
Press to change Speed to Metric Units, Press to return to Imperial Units

Filter Pump Setup:

ON Hour

Select Hour to Automatically turn Filter Pump ON 0-23hr

OFF Hour

Select Hour to Automatically turn Filter Pump OFF 0-23hr

DISABLE – Used for Filter Maintenance

Press to Disable Filter Pump – Button will turn Yellow

Press again to Enable Filter Pump – Button will turn Gray

MANUAL

Press to Select Auto programming for Filter Pump – Button will say AUTO

Press again to return to Manual Mode where Filter Pump runs all the time unless Disabled

REMOTE HMI:

Remote 1 ON

Press to disable USER1 Login for Remote HMI Access

Remote 2 ON

Press to disable USER2 Login for Remote HMI Access

Remote 3 ON

Press to disable USER3 Login for Remote HMI Access

Annunciation:

Treadmill Start Light Warning

Press to enable Light Flash Warning with Treadmill Start

Treadmill Start Beep Warning

Press to enable Beep Warning with Treadmill Start

6.13 Engineering Menu



***Engineering Menu is only used for advanced service and development purposes**

For Questions regarding Installation or Repair of this control panel:

ISTECH Inc.
Ph. (800) 555-4880
Fax. (717) 467-5780
hydroworx@istech.us.com

End of Manual

Resistance Jets

- Starting / Stopping
- Adjusting Flow Rate
- Adjusting Flow Direction
- Pulsating Massage Hose

Starting / Stopping:

1. Activate the desktop pendant by placing the key into the emergency stop button and turning the key to the right (button will pop up).
   ***Nothing in pool will work until pendant is activated.**

2. Push the button marked POWER RESET (button will illuminate when activated).

3. Using either the keypad remote or pendant, press the LEFT BADU button and then push the ON button to turn the power to the jets on (If your pool has two front jets select either LEFT or RIGHT depending on which jet you using).

4. To shut down, press the OFF button.

Adjusting Flow Rate:

1. Press the FAST or SLOW buttons on the keypad remote or pendant to adjust flow rate.

Jets have memory and will always start at the power level previously set when activated.

Adjusting Flow Direction:

1. Grab the jet nozzle and push the nozzle in any direction desired. The nozzle may be stiff and hard to move in the beginning of your use of the HydroWorx Pool, however it will become easier to adjust over time.

Pulsating Massage Hose:

1. Make sure jets are off before connecting.

2. Grab the end of the massage hose with the clip-on device.

3. Pull back on the clip and hold while pushing onto the jet nozzle.

4. When fully inserted onto the jet nozzle, push outside ring of the clip toward the jet to lock massage hose in place.

5. For best results, place the block off jet cap on the opposite jet nozzle. This will force all the water out of the jet nozzle that is attached to the massage hose.

Treadmill System

- Starting / Stopping and Adjusting Speed

Starting / Stopping and Adjusting Speed:

1. Activate the desktop pendant by placing the key into the emergency stop button and turning the key to the right (button will pop up).
   ***Nothing in pool will work until pendant is activated.**

2. Push the button marked POWER RESET (button will illuminate when activated).

3. Using either the keypad remote or pendant, press TREADMILL ON/OFF button to power up the treadmill.

4. Press the FAST or SLOW buttons until desired speed is achieved.

5. To turn off, press the TREADMILL ON/OFF button again (this can be done while the treadmill is running at any speed). The treadmill will gradually slow to a stop.

Treadmill will always reset at 0 mph when turned off and turned back on.

Filtration System

- Changing the Filter Cartridge
- Checking Pump Strainer Baskets
- Checking the Skimmers

Changing the Filter Cartridge:

1. Turn off the power to the heater.
2. Turn off the power to the filter pump.
3. Using the isolation valves, isolate filter system plumbing from pool.
4. Open filter drain.
5. Open the air bleeder valve on top of the filter.
6. After filter is drained, remove clamps around outside of filter.
7. Lift top of filter off and remove cartridge
8. Install new cartridge and put top of filter back on.
9. Reinstall clamps around filter.
10. Close filter drain valve.
11. Return isolation valves back to normal operating position.
12. Air will begin to come out of air bleeder valve on top of filter. Once the air has stopped and water is coming out of the bleeder, close air bleeder valve (If air stops and no water is coming out the filter cartridge is not installed correctly).
13. Turn power on to filter pump.
14. Turn power on to heater.

Checking Pump Strainer Baskets:

1. Turn off the power to the heater.
2. Turn off the power to the filter pump.
3. Using the isolation valves, isolate filter system plumbing from pool.
4. Open the access lid on the pump. Remove basket and clean out any debris. Replace the basket back into the pump. Inspect the O-ring found on the pump lid to be sure it is clean and fully intact. Screw the pump lid firmly back onto pump.
5. Return isolation valves back to normal operating position.
6. Turn power on to filter pump.
7. Turn power on to heater.

Checking the Skimmers:

1. Reach into skimmer housing from inside pool and pull basket out.
2. Clean as necessary.
3. Check skimmers and clean weekly to ensure proper filtration.

Heater

1. Do not operate without water
2. Ensure the heater is properly grounded and bonded
3. Power on the pools circulating system first before powering on the heater
4. Power on the heater
5. Set the temperature to your desired setting

Trouble Shooting:

1. No heat –
   a) Power to the heater not on. Turn on power to heater.
   b) Insufficient water flow through the heater – make sure the circulation pump is operating and the filter is clean. Filter pressure gauge should be in the 10-15 PSI range / if higher the filter needs to be cleaned. Also check bypass valves to make sure enough water is flowing through the heater. Flow light will come on if adequate flow is present.
2. Flow Light ON and Element Light are ON but the unit is not heating water – heating element may need to be replaced.
3. Flow light ON and Element Light OFF – press the reset button

Chemical Control

- Water Chemistry
- Filling Brominator and Adjusting Feed Rate

Water Chemistry:

A Taylor Test Kit is supplied with the purchase of every pool. We recommend reading the Pool & Spa Chemistry book that is supplied with the test kit. It is important to test the pool water, carefully following the test kit instructions, to maintain the proper water chemistry balance. Chemical imbalances in your pool can cause the following: a ring on the shell wall around the water level, green water, and excessive particles sticking to the pool wall. In some cases, prolonged chemistry imbalance can cause damage to the heater, pumps, valves, and seals. This damage would not be covered under the pool warranty. It is also important to keep an accurate daily record of water chemistry. Profiles of your water chemistry components are on the following pages.

WHEN FIRST FILLING YOUR POOL, WE RECOMMEND BALANCING THE WATER CHEMISTRY IMMEDIATELY.

TO DO THIS PROPERLY, FOLLOW THESE STEPS:

1. Check the pH level and set according to the Taylor Test Kit recommendations (7.4 – 7.6 is ideal).
2. Test Alkalinity and Calcium Hardness.
3. Set Alkalinity according to Taylor Test Kit (80 – 120 is ideal).
4. Set Calcium Hardness according to Taylor Test Kit (200 – 400 is ideal).
5. Shock Pool with Lithium Hypochlorite based shock (this chlorine based shock will kill living organisms and bacteria in the water).
6. Test water again later same day and again the next morning.

Typical Supplies Needed to Maintain Pool:

- Alkalinity Increaser
- Calcium Hardness Increaser & Decreaser
- Chlorine based Shock with Calcium Hypochlorite in it.
- pH Decreaser
- Bromine Tablets
- Spa Stain & Scale
- Metal Free
- Pool Perfect
- Garden hose & submersible pump
- (1) Spare pool filter (call HydroWorx to order 717-902-1923)
- Simple Green and non-abrasive sponges (can be picked up at Lowes)
- Test strips (w/ ability to test pH, total alkalinity, calcium hardness, bromine)
- Taylor Test Kit
- Car wax, application cloths and buff pads – taking care of your stainless steel

Filling Brominator:

CAUTION: EYE PROTECTION AND GLOVES ARE REQUIRED. AVOID BREATING FUMES!

1. Close isolation valves before and after Brominator.
2. Carefully remove cover cap.
3. Slowly add 1” Bromine tablets to fill to 1/3 capacity. Try to limit water splashing as much as possible.
4. Replace cover cap securely.
5. Open isolation valves before and after Brominator.

Adjusting Brominator Feed Rate:

1. Upon initial pool fill, set variable setting valve on outside of Brominator to setting #2.
2. After testing water with Taylor Test Kit, adjust Bromine feed rate by increasing or decreasing the variable setting valve.

E-Stop Cable System

- How to Reset E-Stop

Resetting Emergency Stop:

1. It is located on the same side of the pool as the filter pump (close to the control cabinet roughly 5 feet off the floor). It is red and has a cord penetrating the pool wall.

2. To reset, simply push the silver button in.

If this cord has been pulled, no features in the pool will work.

Camera System

The underwater camera system is connected through the DVR and camera view switching device then run to the viewing monitor. You can switch views by selecting them on the camera switcher. Review operation of video switcher/DVR in the manual supplied with the unit.

Draining and Filling the Pool

Draining:

1. Turn off the power to the heater.

2. Turn off the power to the filter pump.

3. Turn the AUTO WATER FILL switch on outside of control panel to the "off" position.

4. Close valve in water line from facility water source to pool.

5. Open drain valves. **Be sure to monitor sump pit or mechanical pit floor drains when draining the pool. If these do not keep up with the draining water flow, partially close the pool drain valves to slow down the water flow.**

6. Raise the treadmill floor to the "Maintenance position". Push the red STOP button on the control pendant to lock the system. Wash down the interior of the pool with a garden hose.

DO NOT PLACE ANY BODY PART BETWEEN THE TREADMILL FLOOR AND THE POOL SIDES AT ANY TIME!

Filling:

1. Close all drain valves.

2. Open valve in water line from facility water source to pool.

3. Turn the AUTO WATER FILL switch on outside of control panel to the "manual" position.

4. Lower the treadmill floor to a level below the skimmers.

5. When pool is full, turn the AUTO WATER FILL switch on outside of control panel to the "auto" position.

6. Bleed out any trapped air inside filter housing (see Filtration System section – Changing the Filter Cartridge).

7. Turn on the power to the filter pump.

8. Turn on the power to the heater.

Pool Maintenance Records

- Stainless Steel Care
- Pool Maintenance Schedule
- Daily Chemical Logs
- Weekly Chemical Logs
- Weekly Maintenance Records
- Monthly Maintenance Records
- Four (4) Month Maintenance Records

Stainless Steel Care:

STAINLESS STEEL CLEANING & MAINTENANCE: THE DO'S AND DON'TS

Stainless steel is a durable, corrosion-resistant chromium/nickel alloy steel with excellent luster. But it's not rustproof—particularly within the harsh environment of a pool. Two issues are at play here: First, chlorine and bromine sanitizers are highly caustic chemicals for stainless steel. Second, heat and humidity enhance the corrosiveness of these chemicals.

Regular cleaning is the best way to prevent corrosion and add to the service life for stainless steel handrails, support bars, treadmill cover plates, exterior decking, and any other pool equipment. The goal of the cleaning & maintenance program should be to keep the protective chromium oxide layer of the stainless steel intact. Only this will prevent corrosion.

The "Do" List: Exposed stainless steel in your HydroWorx Pool (stairway handrails, interior handrails, stainless steel cap around the border of the pool, HydroWorx Logo…

- Clean frequently with a cleaner and water. Any cleaner that is safe for glass is usually safe for stainless steel. **One time per week.**
- Inspect equipment frequently, if you notice discoloration, tarnish or water stains, increase the frequency of your fresh water rinses to reduce accumulated chemicals.
- Remove any rust spots as soon as possible to prevent irreversible pitting. Use a scotch brite pad and always go with the grain.
- **Quick tip for corrosion prevention:** Apply a physical barrier between the stainless steel and corrosive agents by using a soft paste wax, such as an automotive wax. A coating of wax may last for up to six months, depending on equipment usage. You would also want to treat the treadmill cover plate and treadmill support bars. **Do this when you drain and clean the pool.**

The "Don't" List:

- Do not use steel wool or sandpaper, mineral acids, bleaches, or chlorine cleansers.
- Do not add chlorine to the pool in close proximity to stainless steel equipment. Sanitizers should be added as far away as possible from stainless steel.

Pool Maintenance Schedule:

Daily:

- ✓ Check Bromine and pH water chemical levels.

Weekly:

- ✓ Check Bromine tablet level in Brominator.
- ✓ Clean strainer baskets in all pumps.
- ✓ Clean skimmer baskets.
- ✓ Shock Pool with Lithium Hypochlorite Shock.
- ✓ Check Alkalinity and Calcium Hardness water chemical levels.

Monthly:

- ✓ Test the main GFCI breaker.
- ✓ Clean or change the pool filter.

Every Four (4) Months (or as usage dictates):

- ✓ Drain and clean the pool.

Daily Chemical Logs:

Month of_____

Date	Time	pH 7.4 – 7.6	Bromine 3.0 – 5.0	Pool Temp	Clarity	Tester Initials
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						
9.						
10.						
11.						
12.						
13.						
14.						
15.						
16.						
17.						
18.						
19.						
20.						
21.						
22.						
23.						
24.						
25.						
26.						
27.						
28.						
29.						
30.						
31.						

Daily Chemical Logs:

Month of_____

Date	Time	pH 7.4 – 7.6	Bromine 3.0 – 5.0	Pool Temp	Clarity	Tester Initials
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						
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20.						
21.						
22.						
23.						
24.						
25.						
26.						
27.						
28.						
29.						
30.						
31.						

Weekly Chemical Logs:

Date	Time	Alkalinity 80 – 120	Calcium 200 – 400	Pool Temp	Clarity	Tester Initials

Weekly Chemical Logs:

Date	Time	Alkalinity 80 – 120	Calcium 200 – 400	Pool Temp	Clarity	Tester Initials

Weekly Maintenance Records:

Week of_____

TASK	CHECK	INITIAL	DATE
CHECK Bromine Level in Brominator			
CLEAN Strainer Baskets in All Pumps			
CLEAN Skimmer Baskets			
SHOCK Pool with Lithium Hypochlorite			

REMINDER – Pool is not to be used by bathers for 10 hours from when shocked.

Weekly Maintenance Records:

Week of_____

TASK	CHECK	INITIAL	DATE
CHECK Bromine Level in Brominator			
CLEAN Strainer Baskets in All Pumps			
CLEAN Skimmer Baskets			
SHOCK Pool with Lithium Hypochlorite			

REMINDER – Pool is not to be used by bathers for 10 hours from when shocked.

Weekly Maintenance Records:

Week of_____

TASK	CHECK	INITIAL	DATE
CHECK Bromine Level in Brominator			
CLEAN Strainer Baskets in All Pumps			
CLEAN Skimmer Baskets			
SHOCK Pool with Lithium Hypochlorite			

REMINDER – Pool is not to be used by bathers for 10 hours from when shocked.

Weekly Maintenance Records:

Week of_____

TASK	CHECK	INITIAL	DATE
CHECK Bromine Level in Brominator			
CLEAN Strainer Baskets in All Pumps			
CLEAN Skimmer Baskets			
SHOCK Pool with Lithium Hypochlorite			

REMINDER – Pool is not to be used by bathers for 10 hours from when shocked.

Weekly Maintenance Records:

Week of_____

TASK	CHECK	INITIAL	DATE
CHECK Bromine Level in Brominator			
CLEAN Strainer Baskets in All Pumps			
CLEAN Skimmer Baskets			
SHOCK Pool with Lithium Hypochlorite			

REMINDER – Pool is not to be used by bathers for 10 hours from when shocked.

Weekly Maintenance Records:

Week of_____

TASK	CHECK	INITIAL	DATE
CHECK Bromine Level in Brominator			
CLEAN Strainer Baskets in All Pumps			
CLEAN Skimmer Baskets			
SHOCK Pool with Lithium Hypochlorite			

REMINDER – Pool is not to be used by bathers for 10 hours from when shocked.

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Monthly Maintenance Records:

Month of_____

TASK	CHECK	INITIAL	DATE
TEST the Main GFCI Breaker			
CLEAN or CHANGE the Pool Filter			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Four (4) Month Maintenance Records:

Date_____

TASK	CHECK	INITIAL	DATE
DRAIN and CLEAN the Pool			

Installation & Maintenance Inspections
Coates Heaters, Models CE, CPH & PHS.

The following items should be inspection at installation and yearly.

☐ Bond Lug.
The copper lug mounted on the outside of the case. The bond wire shall be solid copper conductor, insulated or bare, not smaller than #8 AWG or rigid metal conduit of brass or other identified corrosion resistant metal.
See NEC 680.26(B)

☐ Ground lug and wire size.
The service ground lug is located within the heater cabinet near the Power Distribution Block (PDB). The ground wire must be insulated copper.
The ground wire must be the same size or larger than the ungrounded conductors. ***See Coates I. O. & M.***

☐ Internal wiring and components.
Inspect internal wires and components at installation for tightness. During transportation the heater is subject to vibration which can cause loose wires and equipment. Yearly inspect the wires for tightness and/or overheating.

☐ Subdivided circuits for heating elements.
All Coates Pool and Spa heaters having a load exceeding 48 Amps have sub-divided circuit protection of not over 60 Amps.
The ampacity of the branch circuit conductors and the rating or setting of overcurrent protective devices shall not be less than 125% of the total nameplate rated load. ***See UL 680.9.***

☐ Ground-fault Circuit Interrupter protection
A Ground-fault Circuit Interrupter (GFCI) is not required if: the heater is 3 phase or rated over 250 Volts or has a load of more than 50 Amps. ***See UL 680.44(B).***

Coates Heaters, Models 12004ST, 12008ST, 12406ST and 12411ST.

☐ 12411ST heaters should not be wired using red wire nuts.

Some Blue wire nuts are rated for 1ea #6 and 1ea #8 stranded wire and some are not. Care should be exercised that only wire nut rated for this use are used. Multi-Tap Connectors such as those made by Ilsco (PCT), T & B (AMT), Burndy (1PLD) or Polaris (IPL) work very well, but are expensive. Brass or copper split nuts can also be used.

☐ Supply Circuit Breaker

Heaters with a current load below 50 Amps must have a GFCI circuit breaker unless the heater is rated 3 phase or over 250V.
The breaker is to be rated at 125% of the rated load.

Model	Load	GFCI Circuit Breaker	Volts	Wire Size
12411ST	46 Amps	60 Amp	240V	#6 @ 75°F
12406ST	23 Amps	30 Amp	240V	#10 @ 75°F
12008ST	40 Amps	50 Amp	208V	#8 @ 75°F
12004ST	20 Amps	25 Amp	208V	#10 @ 75°F

The ground wire must be the same size or larger than the ungrounded conductors.

Electrical Inspection Checklist:

Speck Swim Jet Systems & Equipment Packs

☐ Check for cracks in the jet housing, jet nozzles, and anti-entrapment cover. Do not use if any parts are broken or missing. **Are there any visible cracks? YES or NO. If YES, describe:**

☐ Check plumbing connections at jet pump(s) and maintenance pump for cracks, loose connections, or leaks. **Are there any visible cracks, loose connections or leaks? YES or NO. If YES, describe:**

☐ Check that pumps are installed in a dry, well-ventilated area. Pumps may overheat without proper ventilation. **Are all pumps installed in a dry and well-ventilated area? YES or NO. If NO, describe:**

☐ Check that all pumps are properly bonded and grounded according to the National Electric Code and to your state's electric code. Ground wire should be the same wire gauge or larger than supply (hot) wires, but not smaller than 12 AWG (NEC 680.21 explains connections for pool motors in further detail). Bonding wire should be a minimum of 8 AWG (NEC 680.26 explains bonding in further detail). **Are all pumps properly bonded and grounded according to the National Electric Code and to your state's electric code? YES or NO. If NO, describe:**

☐ Check that there are NO loose electrical connections. **Were there any loose electrical connections found during the inspection?**

Were all identified loose electrical connections corrected? YES or NO. If NO, describe what was not corrected and why:

☐ Check for corrosion at bonding lugs. Replace if necessary. **Were there any bonding lugs that needed to be replaced? YES or NO. If YES, describe:**